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The presence of the Tibetan Plateau lowers atmospheric CO_2 levels via the Atlantic-Pacific carbon seesaw

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ABSTRACT

The presence of the Tibetan Plateau is believed to lower pCO_2^{atm} by stimulating weathering carbon sink, during which the global ocean is considered a passive carbon reservoir despite the tremendous marine carbon inventory. Yet, recent studies reveal that the orographic forcing of the Tibetan Plateau could lead to drastic changes in ocean circulation, which would substantially affect basin-scale carbon storage and hence pCO_2^{atm}. However, this connection between the presence of the Tibetan Plateau and changes in the oceanic carbon inventory remains insufficiently investigated. Here, by employing a state-of-the-art ocean-biogeochemical model, we explore the role of the Tibetan Plateau in determining basin-scale carbon storage patterns based on an idealized experimental design. We find that the presence of the Tibetan Plateau substantially enhances deep Pacific carbon storage and hence lowers pCO_2^{atm} via essential reorganization of the meridional overturning circulation, particularly associated with the development of the Pacific halocline. Moreover, the presence of the Tibetan Plateau greatly affects the oceanic carbon uptake in the Northern Hemisphere, which is likely controlled by the variations in surface alkalinity.

1. Introduction

The presence of the Tibetan Plateau is believed to exert predominant control on the Cenozoic climate–carbon dynamics. The prevailing Uplift Hypothesis (Raymo and Ruddiman, 1992) emphasizes the role of the Tibetan Plateau Uplift (UTP, e.g., Fig. 1a) in influencing atmospheric circulation patterns and enhancing chemical weathering rates. Essentially, increased weathering processes lead to greater riverine alkalinity input into the global ocean, consequently drawing down pCO_2^{atm} and contributing to the Cenozoic cooling. Note that, in this case, the global ocean, holding a tremendous inventory of carbon (~50× atmospheric CO_2) (Friedlingstein et al., 2022) communicating directly with the air, is inexplicitly considered a passive yet well-mixed carbon reservoir responding to the mass balance between the carbon input from deep degassing (e.g., volcanic emission) and weathering carbon sink.

With the aid of numerical models, recent research connects the UTP further with alteration in the oceanic Meridional Overturning Circulation (MOC) (Su et al., 2018; Wen et al., 2022; Yang et al., 2024). A number of model studies indicate that the presence of the Tibetan Plateau could be a pre-requisite for an active Atlantic MOC (AMOC) mode, and the Pacific MOC (PMOC) appears to dominate the thermohaline circulation before the UTP. The model results thereby imply a dramatic switch from a PMOC-dominated ocean to an AMOC-dominated ocean due to the redistribution of surface buoyancy forcing associated with the UTP (Yang et al., 2024), although the phenomenon could vary among models due to differing configurations (Zhang et al., 2022).

The dynamics between the UTP and the altered MOC mode can be (partially) confirmed by geological evidence. Although still debatable, the inferred timing of the major uplift in the Himalayas is no earlier than the Miocene (Song et al., 2024), coinciding with the PMOC–AMOC transition during the Oligocene to Miocene (Ferreira et al., 2018) (Fig. 1b). Moreover, water mass proxies, including Nd isotopes (Thomas,

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2004) and carbon isotopes (Nunes and Norris, 2006), suggest persisted deep-water formation in the North Pacific during the early Cenozoic, while microbial source indicators imply that vigorous North Atlantic Deep Water (NADW) production likely emerged no earlier than the early Oligocene (Wang et al., 2023).

As the MOC distributes freshwater, heat, and crucially, geochemical tracers within the ocean's interior, alteration in the MOC is proven to greatly affect oceanic carbon storage and hence pCO_2^{atm} (e.g., (Du et al., 2022; Yu et al., 2022)). Therefore, the PMOC–AMOC transition associated with the UTP could substantially affect the basin-scale carbon reservoir. However, the connection between the presence of the Tibetan Plateau and the oceanic carbon storage has yet to be validated or tested, marking a clear caveat in the current understanding regarding the climate–carbon dynamics in the Uplift Hypothesis, i.e., to neglect the role of the oceanic carbon cycle in the long-term pCO_2^{atm} change. The caveat is particularly relevant considering the dominant control of the production and recycling of biogenic carbon in the marine realm and ultimately its burial in the sediments on pCO_2^{atm} (Hülse et al., 2017).

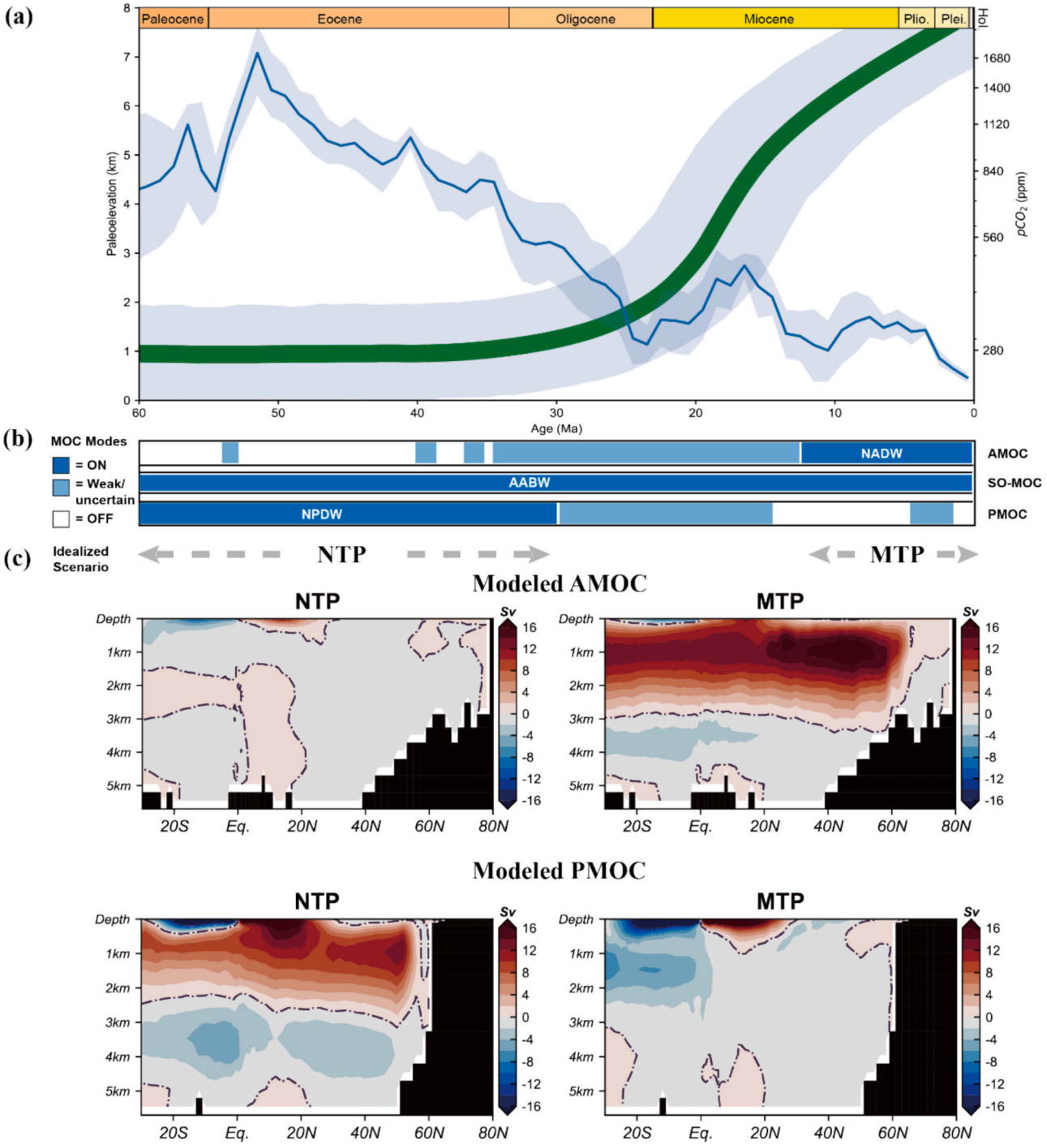


Fig. 1. (a) Paleoaltimetry for the inferred elevation of the Himalaya (adapted from ref. (Spicer et al., 2021)) and long-term pCO_2^{atm} reconstruction from the CenCO2PIP Consortium (Hönisch et al., 2023). (b) Speculated basin-scale MOC modes as in ref. (Ferreira et al., 2018). (c) Modeled AMOC and PMOC in NTP and MTP. Note that the comparison between NTP and MTP only involves changes in the presence of the Tibetan Plateau; other boundary conditions are the same. Therefore, the experimental design is highly idealized and is not intended to represent a realistic scenario constrained by geological evidence.

To this end, we bear speculations that the presence of the Tibetan Plateau may influence basin-scale oceanic carbon storage patterns, thereby potentially contributing to the long-term reduction in pCO_2^{atm}. Here in this study, we leverage oceanic biogeochemical modeling under highly idealized conditions to tentatively validate this hypothesis regarding the presence of the Tibetan Plateau. By analyzing model outputs, we explore how the marine carbon cycle responds to distinct MOC modes based on the analysis of the deep-ocean carbon storage and the factors controlling surface pCO_2.

2. Methods

2.1. Model description

In this study, a state-of-the-art ocean-biogeochemical model is used to investigate the response of the oceanic carbon cycle to the perturbation of the overturning circulations driven by orographic forcing. The physical model is the Massachusetts Institute of Technology ocean general circulation model, MITgcm (Campin et al., 2020; Marshall et al., 1997), and this model is coupled with a dynamic and thermodynamic sea ice module (Losch et al., 2010). In the version used in this study, the Nonlocal K-Profile Parameterization (KPP) scheme is used to consider the turbulent mixing (Large et al., 1994), a bottom layer scheme by Campin and Goose (1999) is incorporated, and the parameterization of sub-grid mesoscale eddies is implemented (Gent and Mcwilliams, 1990). The biogeochemical model used is the Regulated Ecosystem Model 2 (REcoM2, as detailed by (Hauck et al., 2013)), with a globally homogeneous pCO_2^{atm} box (Du et al., 2022) which aims to probe changes in the pCO_2^{atm} and air-sea CO_2 flux. MITgcm-REcoM2 covers a latitude span from 80°S to 80°N. The horizontal resolution is 2 degrees. Vertically, there are 30 unevenly spaced layers, reaching a maximum depth of 5500 m. The biogeochemical REcoM2 contains flexible stoichiometry (Chl:C:N:Si:Fe) during the oceanic primary production based on two different phytoplankton groups, diatoms and small phytoplankton.

MITgcm-REcoM2 has been used to investigate the regional and global oceanic carbon cycle (Hauck et al., 2013; Hauck and Völker, 2015; Pagnone et al., 2019; Ye and Völker, 2017). Specifically, the same configuration of the model has been used to simulate the marine carbon cycle under palaeoceanographic setups such as the Quaternary Glacial–Interglacial cycles (Du et al., 2022).

2.2. Experiment design

The primary aim of this study is to investigate the marine carbon cycle response to the presence of the Tibetan Plateau. Therefore, we conducted and analyzed two experiments, i.e., NTP and MTP. NTP refers to the 'no Tibetan Plateau' conditions, and MTP refers to the 'modern Tibetan Plateau' conditions. As MITgcm-REcoM2 is an ocean–biogeochemical model, we first derived the necessary atmospheric forcing from published experiments (Su et al., 2018) ('input experiments') which used the Community Earth System Model (CESM 1.0.5). The input experiments are referred to as the iNTP and iMTP experiments, respectively.

In the input experiments of iMTP and iNTP, the only difference in the boundary conditions is the topography within an area of 20°–60°N and 60°–140°E (Fig. S1). While modern altitudes in this area are applied in iMTP, every grid point within the same area with an elevation higher than 200 m was set to 200 m in iNTP. Other conditions, including land–sea mask, ocean bathymetry, and orbital parameters, are prescribed to the same pre-industrial conditions. Both iNTP and iMTP were run under $pCO_2^{atm} = 284$ ppm.

Employing atmospheric forcing obtained from iNTP and iMTP, we conducted the two MITgcm-REcoM2 simulations (NTP versus MTP), with initial $pCO_2^{atm} = 284$ ppm. These two experiments aim to elucidate the carbon dynamics responding to the presence of the Tibetan Plateau, particularly associated with the alteration in the MOC system. Additionally, to tentatively investigate the impacts of the UTP under elevated, early Cenozoic-like pCO_2^{atm} levels, we additionally performed three pairs of experiments with increased initial pCO_2^{atm} (560, 840, and 1120 ppm, respectively). Nonetheless, the findings from these three pairs should be considered as only indicative, because the oceanography configuration (whether under MTP or NTP conditions) in these experiments remains the same with the 284-ppm pair. The modeled pCO_2^{atm} are presented in Table S1.

The experiments perfectly align with the idea of validating the effects of UTP directly on the ocean carbon cycle, and the experimental design enables us to examine the effects of the UTP for a generic process understanding in a two-end-member perspective. Yet, as we discussed in the main text, it is important to note that the experimental design neglects several key boundary conditions, including changes in the other mountains, closure of tropical seaways, and changes in the high-latitude ice sheets.

2.3. Forcing fields and Initialization

In the experiments, the MITgcm was driven by climatological forcing fields as derived from the input CESM experiments (Su et al., 2018) iNTP and iMTP, from which we obtained 10-m wind speed, 2-m air temperature, surface latent heat flux, surface sensible heat flux, surface downward longwave and shortwave radiation, evaporation, precipitation, and runoff. As applied previously (Du et al., 2022), we used a 1200-month-long time series to ensure that the forcing fields resolve climate variability on multiple timescales. Sea surface temperature (SST) and sea surface salinity (SSS) were weakly restored (in a time scale of 10 years) to avoid large temperature/salinity drift.

In both MTP and NTP experiments, initial conditions for temperature, salinity, dissolved nitrogen (DIN), and silicate (DSi) were obtained from World Ocean Atlas 2009 (Antonov et al., 2010; Garcia et al., 2010; Locarnini et al., 2010), while alkalinity (ALK) and dissolved inorganic carbon (DIC) were initialized using GLODAP2 data (Olsen et al., 2016). Dissolved iron (DFe) initial values were taken from previous model outputs (Aumont et al., 2015).

Each experiment involved a 3000-year spin-up period considering only physical processes, followed by an additional 2000 years coupled with REcoM2 for biogeochemistry. The biogeochemical model was assumed to have reached quasi-equilibrium after 2000 years, as pCO_2^{atm} variations in the last 100 years are less than 0.2 ppm. The results of the last 100 years were averaged for our analysis.

2.4. Decomposition of the surface pCO_2 signals

In order to enhance our understanding of the factors influencing sea surface pCO_2 (pCO_2^{surf}), we adhered to the methodology previously outlined by (Menviel et al., 2014). We decomposed the variations in MTP-NTP pCO_2^{surf}, into four components: alkalinity (ALK), dissolved inorganic carbon (DIC), sea surface salinity (SSS), and sea surface temperature (SST). Within a theoretical framework, the four variables are expected to affect the solubility of CO_2 in seawater and the partitioning of CO_2 at the air-sea interface (D'Orgeville et al., 2011). Among them, ALK reflects the seawater's capacity to neutralize acids and buffer CO_2 concentrations. An increase in ALK, therefore, tends to lower pCO_2. DIC comprises carbon dioxide (CO_2), bicarbonate (HCO_3^-), and carbonate (CO_3^{2-}) ions. Adding DIC directly increases the concentration of CO_2 in the seawater, which can lead to an increase in pCO_2. Furthermore, an increase of SSS and/or SST typically leads to a decrease in the solubility of CO_2, resulting in an elevation of pCO_2. Here we give an introduction for readability, while a detailed description can be found in the reference (D'Orgeville et al., 2011; Menviel et al., 2014).

$$\Delta pCO_2 = \Delta pCO_{2DIC} + \Delta pCO_{2ALK} + \Delta pCO_{2SSS} + \Delta pCO_{2SST} \quad (1)$$

The first three items on the right-hand side of Eq. (1) are defined as:

$$\Delta pCO_{2X} = \Delta X^{*} \gamma_X {}^{*} pCO_{2Ref} \Big/ \overline{X} \quad (2)$$

Where X represents DIC, or ALK or SSS, the γ_X is a constant coefficient with a value of 10, −9.4, and 1 for DIC, ALK, and SSS, respectively. The last term on the right-hand side of Eq. (1) is defined as:

$$\Delta pCO_{2SST} = exp(\Delta SST^{*} \gamma_{SST}) {}^{*} pCO_{2Ref} - pCO_{2Ref} \quad (3)$$

Where γ_{SST}=0.0423. pCO_{2Ref} is 284 ppm.

3. Results and discussion

3.1. The pacific–atlantic carbon seesaw: oceanic carbon storage under the orographic forcing

The major concern of this study is to investigate the changes in the basin-scale carbon storage pattern driven by the orographic forcing, i.e., the presence of the Tibetan Plateau. For a generic process understanding, we consider the anomaly between the MTP (Modern Tibetan Plateau) and the NTP (no Tibetan Plateau) conditions can to some extent reveal a coarse panorama of the long-term evolution regarding the oceanic carbon cycle responding to the UTP. The MTP and NTP experiments yielded distinct overturning circulation modes (Fig. 1c). Under NTP conditions, vigorous deep-water formation appears in the North Pacific, leading to a well-developed PMOC of ~20 Sverdrups (Sv, 1 Sv = 10^6 m^3/s), reaching a maximum depth of ~3000 m. In contrast, the PMOC is mostly absent in MTP as the AMOC dominates the overturning circulations due to increased net freshwater flux in the North Pacific, as is shown in a previous study (Su et al., 2018).

The distinct MOC modes are then combined with different initial pCO_2^{atm} in the carbon cycle modeling (Methods). Overall, the MTP-NTP anomaly reveals a clear alteration in the carbon cycle. For the pair initialized with 284 ppm pCO_2^{atm}, the modeled pCO_2^{atm} in MTP conditions (284 ppm) is 16 ppm lower than that in NTP conditions (300 ppm), consistent with our speculation that the presence of the TP could facilitate pCO_2^{atm} drawdown (Table S1). In addition, to investigate the magnitude changes of pCO_2^{atm} under higher levels, e.g., the early Cenozoic-like levels, we conducted three paired experiments with elevated initial pCO_2^{atm} at 560, 840, and 1120 ppm (Table S1). Consequently, higher initial pCO_2^{atm} leads to larger MTP-NTP pCO_2^{atm} differences, with more than 100 ppm pCO_2^{atm} change observed in the 1120-ppm pair, corroborating the idea that the NTP to MTP transition may lead to essential oceanic carbon uptake with higher background pCO_2^{atm}, although this is only indicative as the atmospheric forcing applied in the higher pCO_2^{atm} experiments remains the same with the 284-ppm pair. Therefore, for the following analysis of the basin-scale carbon storage, we use the experiments initialized with 284 ppm pCO_2^{atm}.

Our experiments reveal a clear Pacific–Atlantic seesaw pattern in carbon sequestration under contrasting orographic configurations (Table 1). Nearly 500 gigatons of carbon (GtC, 1 GtC = 10^{15} gC) are preferentially sequestered in the Pacific under present-day orography (MTP) compared to the no-Tibetan Plateau scenario (NTP). To understand the processes driving these changes, we analyze the MTP-NTP anomalies in modeled zonal mean dissolved inorganic carbon (DIC, which is a direct tracer of the oceanic carbon storage) concentration and salinity (Fig. 2). Here, the Pacific and Atlantic basins are defined as regions north of 45°S, excluding the Southern Ocean sectors, which will be analyzed separately.

Table 1
Summary of the modeled basin-scale carbon storage changes.**

		NTP	MTP-NTP
Modeled pCO_2^{atm} (ppm)		300	**−16**
	ALL	16,046	+535
Pacific inventory (Gt C)	*Upper (<1000 m)*	3665	+8
	Deep(>1000 m)	12,381	**+527**
	ALL	7637	−280
Atlantic inventory (Gt C)	*Upper (<1000 m)*	1823	−20
	Deep(>1000 m)	5814	−260
Pacific - Atlantic (Gt C)	*ALL*	8409	**+816**
Southern Ocean inventory (Gt C)	*ALL*	6504	−135

** Note that the 'Pacific' and 'Atlantic' are referring to the basins north of the 45 S. The column 'NTP' lists the model results from NTP, while in the column 'MTP-NTP' the anomalies between MTP and NTP are shown. Please notice that with a pCO_2^{atm} reduction of 16 ppm, the difference between the Pacific and Atlantic basins enhances to over 800 GtC.

The Pacific Ocean emerges as the primary sink for the additional carbon sequestered in the MTP ocean (Fig. 2b, left panel). This vast oceanic carbon reservoir expands due to contributions from other parts of the global ocean and the atmosphere (Table 1). However, the increase in carbon storage is not uniform throughout the abyssal Pacific. As shown in Fig. 2b, the abyssal North Pacific exhibits the most significant positive DIC anomalies, exceeding 200 mmol/m^3. These positive anomalies progressively decrease upwards and southward within the Pacific basin. A clear demarcation of the DIC anomaly is evident in the upper 1000 m, marked by a black line in Fig. 2b. The seawater above this line shows negative DIC anomalies concentrated in the upper North Pacific between 30°N and 60°N, leading to a slight carbon loss of the Pacific in the upper 1000 m (Table 1).

Our analysis reveals a prominent role for the halocline in governing the observed asymmetry of the vertical DIC concentration anomaly between the Pacific and Atlantic basins in the experiments. In NTP, characterized by a vigorous PMOC, the surface North Pacific exhibits a higher salinity relative to the North Atlantic (Fig. S2a). This salinity difference facilitates deep convection, leading to a well-mixed layer exceeding 900 m deep on an annual mean basis (Fig. S2b). Consequently, the deeper, well-mixed layer leads to a much weaker DIC vertical gradient (Fig. 2a). Conversely, the MTP scenario (with present-day topography) displays surface freshening in the North Pacific (Fig. S2g), coinciding with the absence of deep convection (Fig. S2h). This fresher upper ocean strengthens the vertical salinity gradient, leading to a pronounced halocline (Fig. 2b). The vertical mixing scheme employed in the model (Large et al., 1994) dictates that diapycnal mixing (that is, mixing across density surfaces) depends on the degree of stratification, which is linked to the vertical salinity gradient. The intensified halocline in the MTP scenario significantly restricts diapycnal mixing, thus limiting carbon exchange between the deep and upper parts of the Pacific Ocean. This restricted exchange ultimately promotes preferential carbon sequestration within the deep Pacific.

The contrasting MOCs induce a mirrored response in the Atlantic's carbon pool. The intermediate to deep Atlantic exhibits pronounced negative DIC concentration anomalies, while the upper ~500 m of the North Atlantic show positive anomalies (Fig. 2b). Moreover, these basin-scale DIC patterns align well with the observed salinity structure. For instance, the negative anomalies coincide with a clear intrusion of saline water, likely reflecting the modern NADW formation process in the MTP scenario. In contrast to the Pacific, the diminished salinity gradient in the Atlantic allows for thorough mixing of the carbon cycle tracers. This efficient mixing leads to the upward transport of deep carbon back to the surface ocean.

In MTP, the Southern Ocean (south of 45°S) is associated with lower DIC concentrations for both Pacific and Atlantic sectors (Fig. 2b). This carbon loss appears to be linked to the varying strengths of the Southern Ocean Meridional Overturning Circulation (SO MOC) in the experiments (Fig. S4). The Southern Ocean acts as a critical pathway for the exchange and modification of water masses within the global thermohaline circulation (Sikes et al., 2023), and a key function of the SO MOC is to bring back the carbon-rich deep water from the abyss to the surface ocean (Yu et al., 2020). A stronger SO MOC would be associated with higher DIC concentrations in the Southern Ocean. In MTP, the SO MOC weakens by about 2 Sv compared to NTP (Fig. S4), resulting in less carbon being transported from the deep ocean to the surface layer of the Southern Ocean. This reduced transport ultimately contributes to the preferential

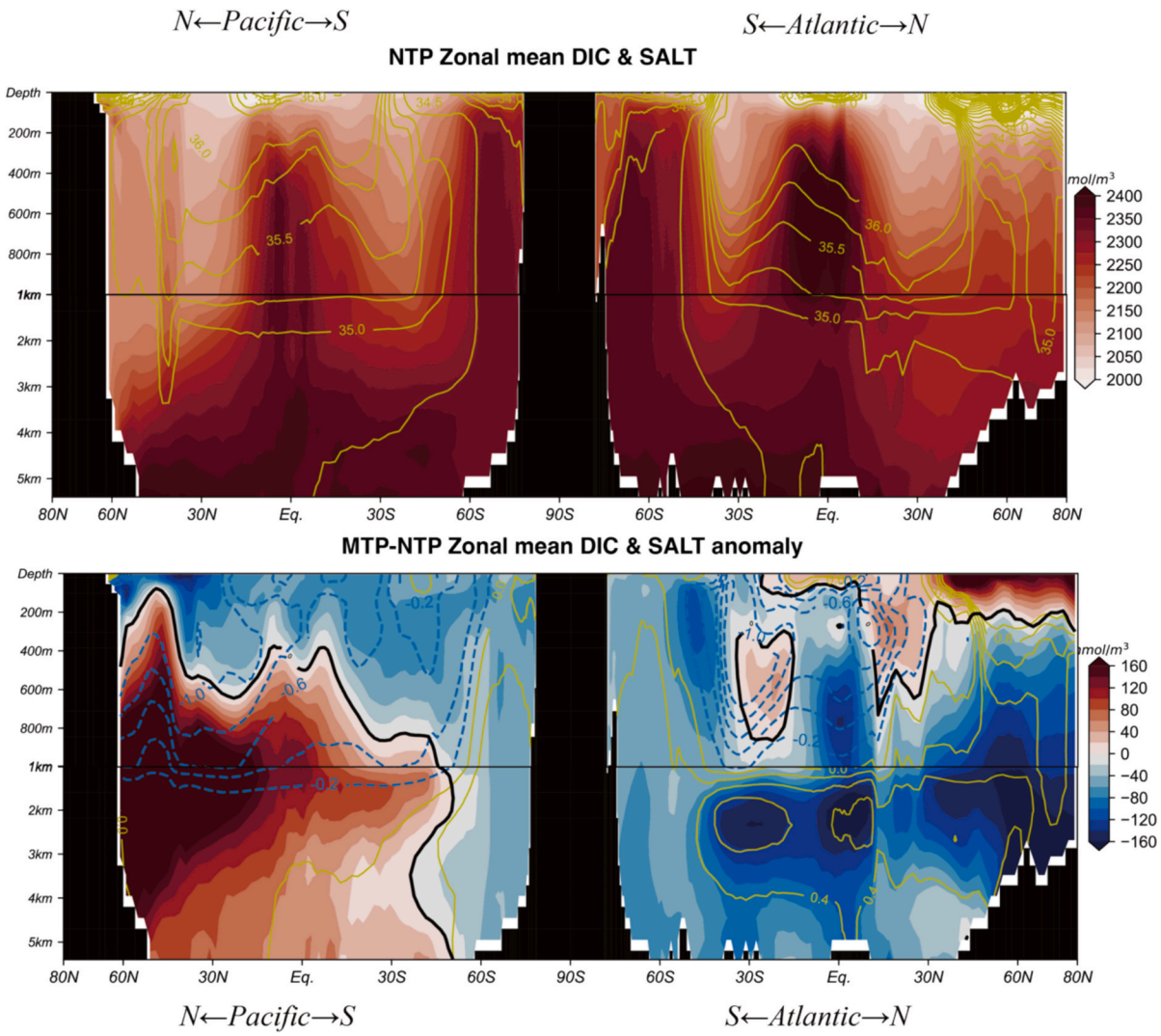


Fig. 2. (a) Zonal mean DIC concentration (shaded) and salinity (contours, in ‰) in NTP. (b) MTP-NTP anomalies of DIC (shaded) and salinity (contours). From the left to the right, the transects run from the Pacific basin to the Atlantic basin, with the Southern Ocean located in the middle. In (b), positive salinity anomalies are marked by golden solid contours, while negative anomalies are in blue dashed lines. The thick, black line in (b) indicates where the DIC anomaly = 0. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

sequestration of carbon within the deep Pacific. Our findings align with previous simulations suggesting a negative correlation between deep Pacific carbon storage and AABW production (Menviel et al., 2014).

3.2. *The underlying mechanism driving the surface dpCO_2 pattern*

Our analysis extends beyond the changes in ocean interior carbon inventory. We additionally illustrate that the substantial shift in overturning circulation patterns triggers a great reorganization of the atmospheric CO_2 source and sink patterns. Fig. 3a illustrates this by depicting the difference in pCO_2 between the ocean surface and the atmosphere (dpCO_2) in NTP. This difference (dpCO_2) ultimately determines the direction and (partially) the magnitude of air–sea CO_2 exchange. In Fig. 3a, negative dpCO_2 values generally indicate a tendency for ocean uptake of CO_2, while positive values indicate outgassing. Although both the North Pacific and North Atlantic operate primarily as atmospheric CO_2 sinks in both the NTP and MTP scenarios (i.e., d$pCO_2 < 0$), the dpCO_2 anomalies (dpCO_2 in MTP - dpCO_2 in NTP) reveal clear interbasin differences (Fig. 3b). The mid-to-high-latitude North Atlantic exhibits negative dpCO_2 anomalies, suggesting that this region becomes a more active carbon sink in the MTP scenario. In contrast, extensive portions of the North Pacific are characterized by positive dpCO_2 anomalies. It is important to note that, in this context, the positive anomalies indicate a reduced CO_2 uptake rather than a transition to outgassing.

At face value, an intriguing disconnection emerges between the surface DIC and dpCO_2 in the North Pacific. In theory, other conditions being unchanged, less CO_2 uptake by the North Pacific should lead to a concurrent decrease in both DIC concentration and the surface pCO_2 (pCO_2^{surf}) (Anderson et al., 2019; Yu et al., 2014) (Method). As in the North Pacific pCO_2^{surf} is lower than pCO_2^{atm}, this means dpCO_2 should decrease (i.e., become more negative). However, the model simulates a rise in dpCO_2 (less negative) accompanied by a decrease in DIC concentration within the North Pacific, and the converse is observed in the North Atlantic (decreasing dpCO_2 with increasing DIC). This highlights a more complex relationship between dpCO_2 and its controlling environmental factors.

By decomposing the dpCO_2 signals (Method), our analysis underscores that changes in ALK exert dominant influences on the surface pCO_2 differences between the MTP and NTP scenarios across the Northern Hemisphere. The effects induced by changes in ALK are counterbalanced by the variations in DIC, while changes in SSS and SST may play a minor role.

Here we take the North Pacific as an instance. When contrasting to

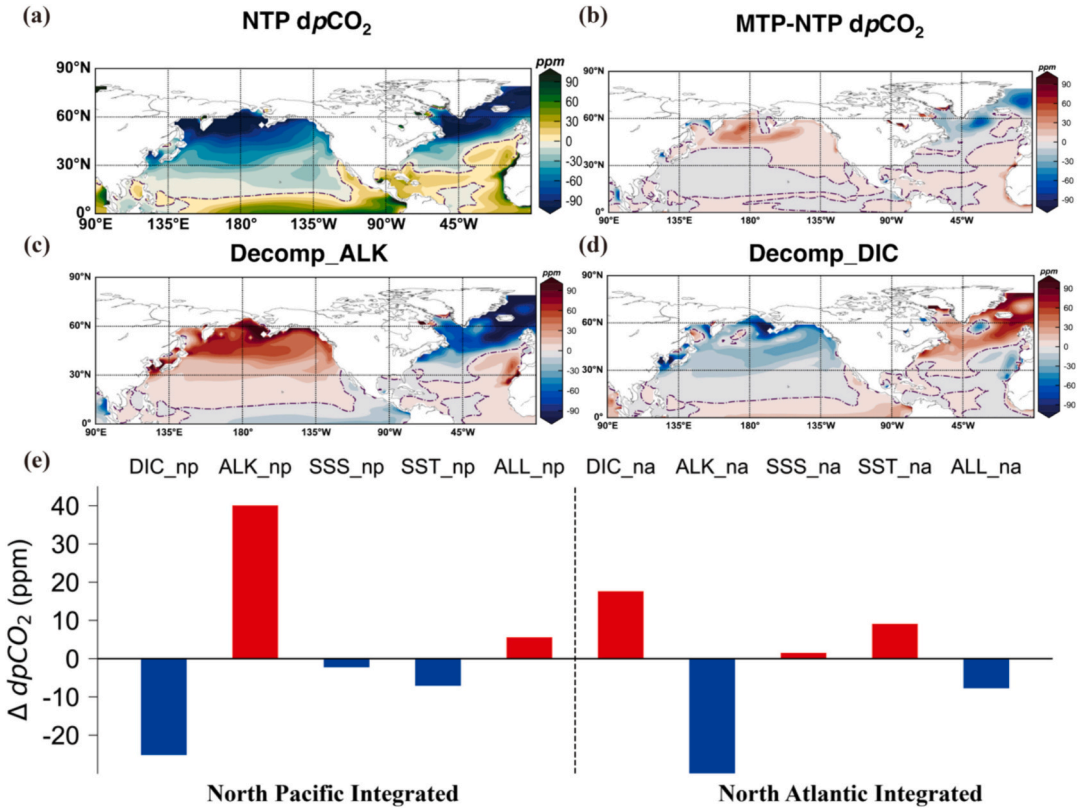


Fig. 3. (a) The modeled dpCO_2 (pCO_2^{surf}-pCO_2^{atm}) in NTP. (b) MTP-NTP dpCO_2 anomalies. In (c)(d), ALK and DIC contribution to the dpCO_2 anomalies are shown. (e) Integrated basin-specific dpCO_2 anomalies decomposition.

NTP conditions, the North Pacific in MTP exhibits a decrease in ALK (Fig. S5), DIC (Fig. 2b), SSS (Fig. S2g), and SST (Fig. S2i). The reduction of the surface DIC concentration (Fig. 2b) tends to lower pCO_2^{surf}, hence promoting CO_2 uptake (Fig. 3e). However, the presence of the North Pacific halocline not only restricts DIC exchanges between the deep and upper waters but also leads to a substantial decrease in upper-Pacific ALK (Fig. S5). The clear reduction in surface ALK (Fig. S5) in the MTP scenario pushes pCO_2^{surf} upwards (Fig. 3e), even surpassing the effect of the DIC. Overall, integrated North Pacific pCO_2^{surf} change induced by the ALK reduction is about +40 ppm, which is clearly beyond the magnitude of that induced by the DIC reduction (−25 ppm). Meanwhile, the substantial positive influence of ALK on pCO_2^{surf} is partially offset by the decline of SSS (−2 ppm) and SST (−7 ppm) of the surface North Pacific, although these effects are minor. Combining these four contributors results in moderate positive dpCO_2 anomalies in the North Pacific. The North Atlantic exhibits a mirrored response with opposite signs, such as the positive effect of increasing ALK (lowering pCO_2^{surf} by −36 ppm) being counterbalanced by increasing DIC (+17 ppm), warming of the surface ocean (+9 ppm), and increasing salinity (+1 ppm), leaving the North Atlantic more active in carbon uptake.

3.3. Integration with previous studies

While the highly idealized configuration of the experiments should not be considered a realistic representation of the full course of the Cenozoic pCO_2^{atm} change, the presented mechanisms are nevertheless in line with previous carbon cycle model studies. For instance, on millennial time scales, Menviel et al. (2012, 2014) showed that alteration in the MOC leads to clear reorganization of surface pCO_2 pattern determined by the interaction between the DIC and ALK, with DIC in their model playing a more important role. In addition, the feature of high DIC concentration in the NTP upper Pacific is comparable with a recently published study based on the mid-Pliocene boundary conditions (Ford et al., 2022) with a fine-tuned active PMOC mode.

More importantly, our results corroborate with proxy-based reconstruction and calculation. With the aid of foraminiferal $\delta^{11}B$, Shankle et al. (2021) suggested that the upper Pacific is more acidic (i.e., with a lower pH) at the mid-Pliocene associated with active North Pacific Deep Water production. This agrees with our results that the upper Pacific in NTP is associated with less ALK. In the long term, the calculation by Zeebe and Tyrrell (2019) showed a clear long-term increase (~100 mmol/m^3 throughout the Cenozoic) of surface carbonate ion concentration [CO_3^{2-}], associated with the long-term transition between PMOC and AMOC (Fig. 4). As the dataset in their study is to some extent concentrated in the Atlantic only, we plotted the modeled Atlantic [CO_3^{2-}] in Fig. 4 and found that the results agree well with the calculation. The consistency between our results and the independent proxy-based studies addresses the robustness of the model configuration. Further, these results point to the conclusion that the global ocean may not merely be a passive carbon reservoir responding to long-term mass balance of deep-degassing and chemical weathering but is capable of triggering substantial pCO_2^{atm} change on a timescale of several thousand years, which could trigger profound climate feedback.

It is important to acknowledge that the idealized experimental design used in our experiments intentionally neglects the changes in several other key boundary conditions. These include the closure state of major seaways, which can substantially influence the overturning system (Hu et al., 2012; Liu and Hu, 2015; Tan et al., 2022; Toggweiler and Samuels, 1995); the uplift history of the American mountain ranges, impacting atmospheric circulation and the freshwater budget between the Pacific and Atlantic basins (Seager et al., 2002; Sepulchre et al., 2014); the background pCO_2^{atm} itself (in the input climate model runs), which largely influences the surface heat budget and hence the climatology (Foster et al., 2017); the polar ice sheets, which is known to shape the climate system during the mid- to late-Cenozoic era (Knorr and Lohmann, 2014); and the mysterious long-term variation in the ocean

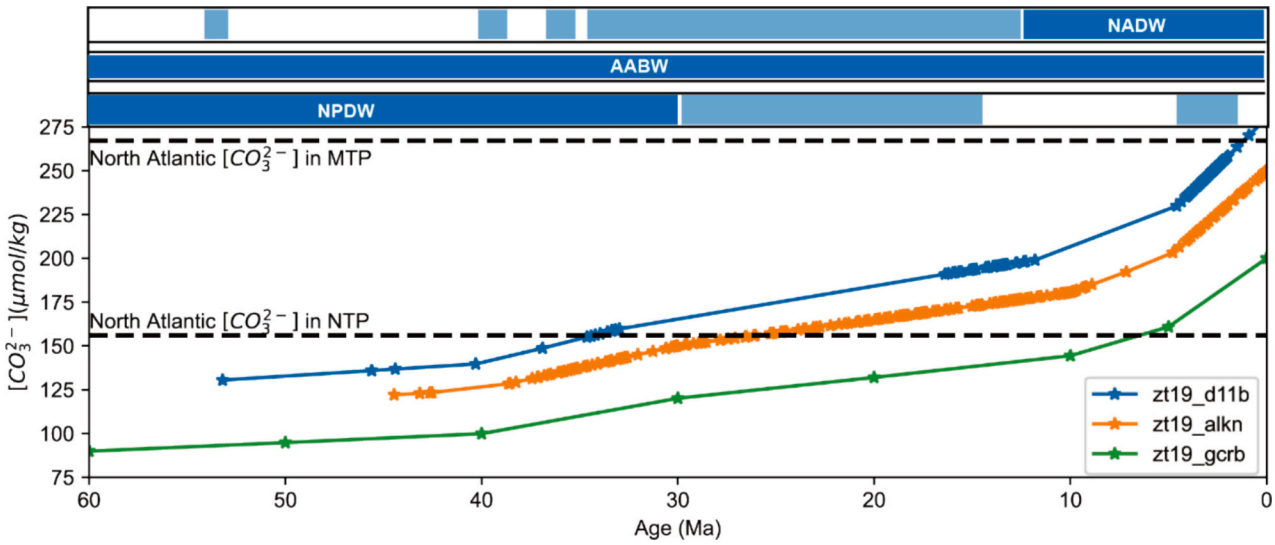


Fig. 4. Comparison between the modeled surface carbonate ion concentration [CO_3^{2-}] with the long-term reconstruction of surface [CO_3^{2-}] from ref.(Zeebe and Tyrrell, 2019). In the plot, 'zt19_xxx' marks the [CO_3^{2-}] under different scenarios: 'gcrb' refers to the GEOCARB model(Berner and Kothavala, 2001), 'd11B' refers to the foraminiferal $\delta^{11}B$ scenario, and 'alk' refers to the alkenone scenario.

alkalinity (Caves et al., 2016), which controls the partitioning of carbon between the atmosphere and the global ocean. The effects of these additional factors on the long-term evolution of the marine carbon cycle are yet to be fully investigated.

Moreover, it is crucial to recognize that the direct influence of the Tibetan Plateau on the proposed PMOC-AMOC transition remains an unresolved issue, which may, to a certain degree, complicate the connection between the UTP and the accumulation of the Pacific carbon inventory. In a recent study focused on the MOC during the early Eocene (47–56 Ma, prior to the presence of the Tibetan Plateau), eight coupled climate model simulations from the Deep-Time Model Intercomparison Project (DeepMIP) were analyzed. The findings revealed that only two out of the eight models produced moderate or vigorous PMOC (Zhang et al., 2022). Absence of the PMOC in the models appears to contradict the proxies (such as Nd and carbon isotopic data) implying deep-water formation in the subpolar North Pacific during the early Cenozoic (Nunes and Norris, 2006; Thomas, 2004), indicating inconsistencies between current models and proxies. While further research is needed to resolve the discrepancies, our simulations reveal that the diminished PMOC could provide a crucial mechanism to enhance Pacific carbon

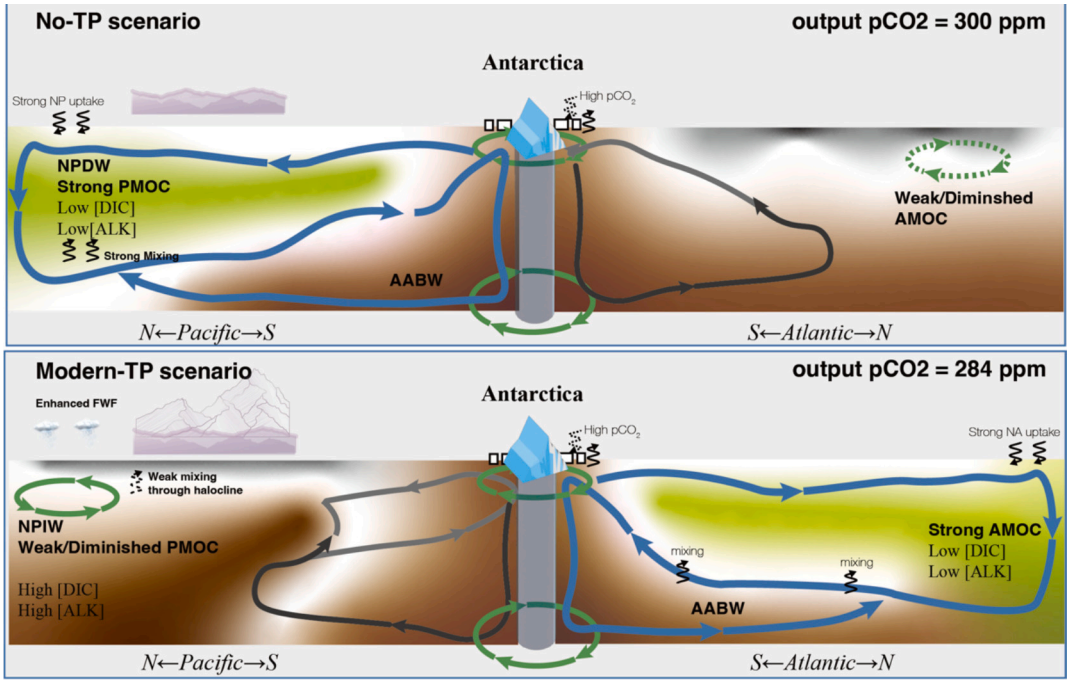


Fig. 5. Conceptual diagram for the MTP-NTP comparison. The thick, solid arrows roughly depict the thermohaline circulation pattern in the Pacific and the Atlantic basin. The concentration of the carbon cycle tracers (DIC, ALK) is shaded by different colors. Under the no-TP scenario, the active PMOC is associated with strong diapycnal mixing in the Pacific, featuring rather low DIC concentration. In comparison, under the modern-TP scenario, the North Pacific becomes the endpoint of the thermohaline circulation with the modern North Pacific Intermediate Water production. The MTP condition introduces the well-developed halocline in the Pacific, promoting carbon sequestration in the deep Pacific.

storage and thereby contribute to the long-term pCO_2^{atm} drawdown.

4. Conclusion remarks and implications

Our findings demonstrate that the presence of the Tibetan Plateau could trigger sophisticated alterations to the marine carbon cycle, impacting both deep-ocean carbon storage (Fig. 2) and surface pCO_2 pattern (Fig. 3), ultimately leading to a reduction in pCO_2^{atm}, as depicted by a conceptual diagram in Fig. 5. Specially, whenever the initial pCO_2^{atm} is elevated, we observe more substantial drops in pCO_2^{atm} (Table S1). Compared with the NTP scenario, the modern ocean (MTP) features a substantial isolation of the deep Pacific with diminishing or even vanished deep-water production. In this case, the presence of the halocline prevents carbon leakage from the deep Pacific, promoting carbon sequestration, and ultimately in favor of lower pCO_2^{atm}. Therefore, our results indicate that under the orographic forcing of the Tibetan Plateau, the global ocean itself appears to play a role in regulating pCO_2^{atm} levels, challenging the idea that the ocean could be merely a passive carbon reservoir in the long term. The close connection among the three factors—the presence of the Tibetan Plateau, alteration in the MOC, and reorganization of basin-scale carbon sequestration—adds an intriguing facet to the context of the Uplift Hypothesis by integrating the effects of the oceanic carbon reservoir into consideration.

In addition, the modeled carbon storage and surface $dpCO_2$ pattern in NTP may show significant implications for future climate–carbon dynamics. For instance, a recent study investigating the $4xCO_2$ scenario results in the activation of the PMOC (Curtis and Fedorov, 2024) and predicts a weakening or disappearance of the Pacific halocline. Under the elevated, similar initial pCO_2^{atm} setup (1120 ppm, Table S1), our model yields more than 100 ppm increase with the active PMOC mode. Therefore, our findings suggest that this (re-)ventilation of the North Pacific could potentially trigger deep carbon leakage from the Pacific Ocean. This carbon leakage, possibly on the magnitude of ~100 GtC as shown in our results, is neglected in the current understanding of climate–carbon dynamics, and further calls for the urgency of mitigating anthropogenic carbon emissions and developing strategies to preserve the ocean's capacity to sequester carbon.

CRediT authorship contribution statement

Jinlong Du: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Jun Tian:** Writing – original draft, Project administration, Funding acquisition, Conceptualization. **Aixue Hu:** Writing – original draft, Visualization, Conceptualization. **Yongqiang Yu:** Writing – original draft, Conceptualization. **Baohuang Su:** Writing – review & editing, Data curation. **Dabang Jiang:** Writing – review & editing, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.gloplacha.2024.104681.

Data availability

Data will be made available on request.

References

Anderson, R.F., Sachs, J.P., Fleisher, M.Q., Allen, K.A., Yu, J., Koutavas, A., Jaccard, S.L., 2019. Deep-Sea Oxygen Depletion and Ocean Carbon Sequestration during the last Ice Age. Glob. Biogeochem. Cycles 33, 301–317. https://doi.org/10.1029/2018GB006049.

Aumont, O., Ethé, C., Tagliabue, A., Bopp, L., Gehlen, M., 2015. PISCES-v2: an ocean biogeochemical model for carbon and ecosystem studies. Geosci. Model Dev. 8, 2465–2513. https://doi.org/10.5194/gmd-8-2465-2015.

Berner, R.A., Kothavala, Z., 2001. Geocarb III: a revised model of atmospheric CO2 over phanerozoic time. Am. J. Sci. 301, 182–204. https://doi.org/10.2475/ajs.301.2.182.

Campin, J.-M., Goose, H., 1999. Parameterization of density-driven downsloping flow for a coarse-resolution ocean model in z-coordinate. Tellus A 51, 412–430. https://doi.org/10.1034/j.1600-0870.1999.t01-3-00006.x.

Campin, J.-M., Heimbach, P., Losch, M., Forget, G., Edhill, Adcroft, A., Amolod Menemenlis, D., Hill, C., Jahn, O., Scott, J., Stephdut Mazloff, M., Fox-Kemper, B., Antnguyen Doddridge, E., Fenty, I., Bates, M., AndrewEichmann-NOAA, Smith, T., Martin, T., Lauderdale, J., Abernathey, R., Samarkhatiwala, Hongandyan, Deremble, B., Dngoldberg Bourgault, P., Dussin, Raphael, 2020. MITgcm/MITgcm: mid 2020 version. https://doi.org/10.5281/ZENODO.3967889.

Caves, J.K., Jost, A.B., Lau, K.V., Maher, K., 2016. Cenozoic carbon cycle imbalances and a variable weathering feedback. Earth Planet. Sci. Lett. 450, 152–163. https://doi.org/10.1016/j.epsl.2016.06.035.

Curtis, P.E., Fedorov, A.V., 2024. Spontaneous Activation of the Pacific Meridional Overturning Circulation (PMOC) in Long-Term Ocean Response to Greenhouse Forcing. J. Clim. 37, 1551–1565. https://doi.org/10.1175/JCLI-D-23-0393.1.

D'Orgeville, M., England, M.H., Sijp, W.P., 2011. Buffered versus non-buffered ocean carbon reservoir variations: Application to the sensitivity of atmospheric pCO2 to ocean circulation changes. Geophys. Res. Lett. 38, 1–5. https://doi.org/10.1029/2011GL049823.

Du, J., Ye, Y., Zhang, X., Völker, C., Tian, J., 2022. Southern Control of Interhemispheric Synergy on Glacial Marine Carbon Sequestration. Geophys. Res. Lett. 49, 1–12. https://doi.org/10.1029/2022GL099048.

Ferreira, D., Cessi, P., Coxall, H.K., de Boer, A., Dijkstra, H.A., Drijfhout, S.S., Eldevik, T., Harnik, N., McManus, J.F., Marshall, D.P., Nilsson, J., Roquet, F., Schneider, T., Wills, R.C., 2018. Atlantic-Pacific Asymmetry in Deep Water Formation. Annu. Rev. Earth Planet. Sci. 46, 327–352. https://doi.org/10.1146/annurev-earth-082517-010045.

Ford, H.L., Burls, N.J., Jacobs, P., Jahn, A., Caballero-Gill, R.P., Hodell, D.A., Fedorov, A. V., 2022. Sustained mid-Pliocene warmth led to deep water formation in the North Pacific. Nat. Geosci. 15, 658–663. https://doi.org/10.1038/s41561-022-00978-3.

Foster, G.L., Royer, D.L., Lunt, D.J., 2017. Future climate forcing potentially without precedent in the last 420 million years. Nat. Commun. 8, 14845. https://doi.org/10.1038/ncomms14845.

Friedlingstein, P., O'Sullivan, M., Jones, M.W., Andrew, R.M., Gregor, L., Hauck, J., Le Quéré, C., Luijkx, I.T., Olsen, A., Peters, G.P., Peters, W., Pongratz, J., Schwingshackl, C., Sitch, S., Canadell, J.G., Ciais, P., Jackson, R.B., Alin, S.R., Alkama, R., Arneth, A., Arora, V.K., Bates, N.R., Becker, M., Bellouin, N., Bittig, H.C., Bopp, L., Chevallier, F., Chini, L.P., Cronin, M., Evans, W., Falk, S., Feely, R.A., Gasser, T., Gehlen, M., Gkritzalis, T., Gloege, L., Grassi, G., Gruber, N., Gürses, Ö., Harris, I., Hefner, M., Houghton, R.A., Hurtt, G.C., Iida, Y., Ilyina, T., Jain, A.K., Jersild, A., Kadono, K., Kato, E., Kennedy, D., Klein Goldewijk, K., Knauer, J., Korsbakken, J.I., Landschützer, P., Lefèvre, N., Lindsay, K., Liu, J., Liu, Z., Marland, G., Mayot, N., McGrath, M.J., Metzl, N., Monacci, N.M., Munro, D.R., Nakaoka, S.-I., Niwa, Y., O'Brien, K., Ono, T., Palmer, P.I., Pan, N., Pierrot, D., Pocock, K., Poulter, B., Resplandy, L., Robertson, E., Rödenbeck, C., Rodriguez, C., Rosan, T.M., Schwinger, J., Séférian, R., Shutler, J.D., Skjelvan, I., Steinhoff, T., Sun, Q., Sutton, A.J., Sweeney, C., Takao, S., Tanhua, T., Tans, P.P., Tian, X., Tian, H., Tilbrook, B., Tsujino, H., Tubiello, F., van der Werf, G.R., Walker, A.P., Wanninkhof, R., Whitehead, C., Willstrand Wranne, A., Wright, R., Yuan, W., Yue, C., Yue, X., Zaehle, S., Zeng, J., Zheng, B., 2022. Global Carbon Budget 2022. Earth Syst. Sci. Data 14, 4811–4900. https://doi.org/10.5194/essd-14-4811-2022.

Gent, P.R., Mcwilliams, J.C., 1990. Isopycnal Mixing in Ocean Circulation Models. J. Phys. Oceanogr. 20, 150–155. https://doi.org/10.1175/1520-0485(1990)020<0150:IMIOCM>2.0.CO;2.

Hauck, J., Völker, C., 2015. Rising atmospheric CO 2 leads to large impact of biology on Southern Ocean CO 2 uptake via changes of the Revelle factor. Geophys. Res. Lett. 42, 1459–1464. https://doi.org/10.1002/2015GL063070.

Hauck, J., Völker, C., Wang, T., Hoppema, M., Losch, M., Wolf-Gladrow, D.A., 2013. Seasonally different carbon flux changes in the Southern Ocean in response to the southern annular mode. Glob. Biogeochem. Cycles 27, 1236–1245. https://doi.org/10.1002/2013GB004600.

Hönisch, B., Royer, D.L., Breecker, D.O., Polissar, P.J., Bowen, G.J., Henehan, M.J., Cui, Y., Steinthorsdottir, M., McElwain, J.C., Kohn, M.J., Pearson, A., Phelps, S.R.,

Uno, K.T., Ridgwell, A., Anagnostou, E., Austermann, J., Badger, M.P.S., Barclay, R. S., Bijl, P.K., Chalk, T.B., Scotese, C.R., de la Vega, E., DeConto, R.M., Dyez, K.A., Ferrini, V., Franks, P.J., Giulivi, C.F., Gutjahr, M., Harper, D.T., Haynes, L.L., Huber, M., Snell, K.E., Keisling, B.A., Konrad, W., Lowenstein, T.K., Malinverno, A., Guillermic, M., Mejía, L.M., Milligan, J.N., Morton, J.J., Nordt, L., Whiteford, R., Roth-Nebelsick, A., Rugenstein, J.K.C., Schaller, M.F., Sheldon, N.D., Sosdian, S., Wilkes, E.B., Witkowski, C.R., Zhang, Y.G., Anderson, L., Beerling, D.J., Bolton, C., Cerling, T.E., Cotton, J.M., Da, J., Ekart, D.D., Foster, G.L., Greenwood, D.R., Hyland, E.G., Jagniecki, E.A., Jasper, J.P., Kowalczyk, J.B., Kunzmann, L., Kürschner, W.M., Lawrence, C.E., Lear, C.H., Martínez-Botí, M.A., Maxbauer, D.P., Montagna, P., Naafs, B.D.A., Rae, J.W.B., Raitzsch, M., Retallack, G.J., Ring, S.J., Seki, O., Sepúlveda, J., Sinha, A., Tesfamichael, T.F., Tripati, A., van der Burgh, J., Yu, J., Zachos, J.C., Zhang, L., 2023. Toward a Cenozoic history of atmospheric CO 2. Science (80-.) 382. https://doi.org/10.1126/science.adi5177.

Hu, A., Meehl, G.A., Han, W., Abe-Ouchi, A., Morrill, C., Okazaki, Y., Chikamoto, M.O., 2012. The Pacific-Atlantic seesaw and the Bering Strait. Geophys. Res. Lett. 39, 1–6. https://doi.org/10.1029/2011GL050567.

Hülse, D., Arndt, S., Wilson, J.D., Munhoven, G., Ridgwell, A., 2017. Understanding the causes and consequences of past marine carbon cycling variability through models. Earth-Sci. Rev. 171, 349–382. https://doi.org/10.1016/j.earscirev.2017.06.004.

Knorr, G., Lohmann, G., 2014. Climate warming during antarctic ice sheet expansion at the middle miocene transition. Nat. Geosci. 7, 376–381. https://doi.org/10.1038/ngeo2119.

Large, W.G., McWilliams, J.C., Doney, S.C., 1994. Oceanic vertical mixing: a review and a model with a nonlocal boundary layer parameterization. Rev. Geophys. 32, 363. https://doi.org/10.1029/94RG01872.

Liu, W., Hu, A., 2015. The role of the PMOC in modulating the deglacial shift of the ITCZ. Clim. Dyn. 45, 3019–3034. https://doi.org/10.1007/s00382-015-2520-6.

Losch, M., Menemenlis, D., Campin, J.-M., Heimbach, P., Hill, C., 2010. On the formulation of sea-ice models. Part 1: Effects of different solver implementations and parameterizations. Ocean Model 33, 129–144. https://doi.org/10.1016/j.ocemod.2009.12.008.

Marshall, J., Hill, C., Perelman, L., Adcroft, A., 1997. Hydrostatic, quasi-hydrostatic, and nonhydrostatic ocean modeling. J. Geophys. Res. Ocean. 102, 5733–5752. https://doi.org/10.1029/96JC02776.

Menviel, L., et al., 2012. Removing the North Pacific halocline: Effects on global climate, ocean circulation and the carbon cycle. Deep. Res. Part II Top. Stud. Oceanogr. 61–64, 106–113.

Menviel, L., England, M.H., Meissner, K.J., Mouchet, A., Yu, J., 2014. Atlantic-Pacific seesaw and its role in outgassing CO2 during Heinrich events. Paleoceanography 29, 58–70. https://doi.org/10.1002/2013PA002542.

Nunes, F., Norris, R.D., 2006. Abrupt reversal in ocean overturning during the Palaeocene/Eocene warm period. Nature 439, 60–63. https://doi.org/10.1038/nature04386.

Olsen, A., Key, R.M., van Heuven, S., Lauvset, S.K., Velo, A., Lin, X., Schirnick, C., Kozyr, A., Tanhua, T., Hoppema, M., Jutterström, S., Steinfeldt, R., Jeansson, E., Ishii, M., Pérez, F.F., Suzuki, T., 2016. The Global Ocean Data Analysis Project version 2 (GLODAPv2) – an internally consistent data product for the world ocean. Earth Syst. Sci. Data 8, 297–323. https://doi.org/10.5194/essd-8-297-2016.

Pagnone, A., Völker, C., Ye, Y., 2019. Processes affecting dissolved iron across the Subtropical North Atlantic: a model study. Ocean Dyn. 69, 989–1007. https://doi.org/10.1007/s10236-019-01288-w.

Raymo, M.E., Ruddiman, W.F., 1992. Tectonic forcing of late Cenozoic climate. Nature 359, 117–122. https://doi.org/10.1038/359117a0.

Seager, R., Battisti, D.S., Yin, J., Gordon, N., Naik, N., Clement, A.C., Cane, M.A., 2002. Is the Gulf Stream responsible for Europe's mild winters? Q. J. R. Meteorol. Soc. 128, 2563–2586. https://doi.org/10.1256/qj.01.128.

Sepulchre, P., Arsouze, T., Donnadieu, Y., Dutay, J.C., Jaramillo, C., Le Bras, J., Martin, E., Montes, C., Waite, A.J., 2014. Consequences of shoaling of the central American Seaway determined from modeling Nd isotopes. Paleoceanography 29, 176–189. https://doi.org/10.1002/2013PA002501.

Shankle, M.G., Burls, N.J., Fedorov, A.V., Thomas, M.D., Liu, W., Penman, D.E., Ford, H. L., Jacobs, P.H., Planavsky, N.J., Hull, P.M., 2021. Pliocene decoupling of equatorial Pacific temperature and pH gradients. Nature 598, 457–461. https://doi.org/10.1038/s41586-021-03884-7.

Sikes, E.L., Umling, N.E., Allen, K.A., Ninnemann, U.S., Robinson, R.S., Russell, J.L., Williams, T.J., 2023. Southern Ocean glacial conditions and their influence on deglacial events. Nat. Rev. Earth Environ. 4, 454–470. https://doi.org/10.1038/s43017-023-00436-7.

Song, Z., Wan, S., Yu, Z., Yu, M., Colin, C., Tang, Y., Zhang, J., Jin, H., Zhao, D., Shi, X., Li, A., 2024. The major uplift in Himalayas was no earlier than the Miocene: evidence from marine sediment record in the Bay of Bengal. Palaeogeogr. Palaeoclimatol. Palaeoecol. 648, 112275. https://doi.org/10.1016/j.palaeo.2024.112275.

Spicer, R.A., Su, T., Valdes, P.J., Farnsworth, A., Wu, F.-X., Shi, G., Spicer, T.E.V., Zhou, Z., 2021. Why 'the uplift of the Tibetan Plateau' is a myth. Natl. Sci. Rev. 8. https://doi.org/10.1093/nsr/nwaa091.

Su, B., Jiang, D., Zhang, R., Sepulchre, P., Ramstein, G., 2018. Difference between the North Atlantic and Pacific meridional overturning circulation in response to the uplift of the Tibetan Plateau. Clim. Past 14, 751–762. https://doi.org/10.5194/cp-14-751-2018.

Tan, N., Zhang, Z.S., Guo, Z.T., Guo, C.C., Zhang, Z.J., He, Z.L., Ramstein, G., 2022. Recognizing the Role of Tropical Seaways in Modulating the Pacific Circulation. Geophys. Res. Lett. 49, 1–10. https://doi.org/10.1029/2022GL099674.

Thomas, D.J., 2004. Evidence for deep-water production in the North Pacific Ocean during the early Cenozoic warm interval. Nature 430, 65–68. https://doi.org/10.1038/nature02639.

Toggweiler, J.R., Samuels, B., 1995. Effect of drake passage on the global thermohaline circulation. Deep Sea Res. Part I Oceanogr. Res. Pap. 42, 477–500. https://doi.org/10.1016/0967-0637(95)00012-U.

Wang, H., Liu, W., Lu, H., Zhang, Y., Liang, Y., He, Y., Bohaty, S.M., Wilson, P.A., Liu, Z., 2023. Oxygenated deep waters fed early Atlantic overturning circulation upon Antarctic glaciation. Nat. Geosci. 16, 1014–1019. https://doi.org/10.1038/s41561-023-01292-2.

Wen, Q., Yang, H., Yang, K., Li, G., Liu, Z., Liu, J., 2022. Possible thermal effect of Tibetan Plateau on the Atlantic Meridional Overturning Circulation. Geophys. Res. Lett. 49, 1–11. https://doi.org/10.1029/2021GL095771.

Yang, H., Jiang, R., Wen, Q., Liu, Y., Wu, G., Huang, J., 2024. The role of mountains in shaping the global meridional overturning circulation. Nat. Commun. 15, 2602. https://doi.org/10.1038/s41467-024-46856-x.

Ye, Y., Völker, C., 2017. On the Role of Dust-Deposited Lithogenic Particles for Iron Cycling in the Tropical and Subtropical Atlantic. Glob. Biogeochem. Cycles 31, 1543–1558. https://doi.org/10.1002/2017GB005663.

Yu, J., Anderson, R.F., Rohling, E.J., 2014. Deep ocean carbonate chemistry and glacial-interglacial atmospheric co2 changes. Oceanography 27, 16–25. https://doi.org/10.5670/oceanog.2014.04.

Yu, J., Menviel, L., Jin, Z.D., Anderson, R.F., Jian, Z., Piotrowski, A.M., Ma, X., Rohling, E.J., Zhang, F., Marino, G., McManus, J.F., 2020. Last glacial atmospheric CO2 decline due to widespread Pacific deep-water expansion. Nat. Geosci. 13, 628–633. https://doi.org/10.1038/s41561-020-0610-5.

Yu, J., Oppo, D.W., Jin, Z., Lacerra, M., Ji, X., Umling, N.E., Lund, D.C., McCave, N., Menviel, L., Shao, J., Xu, C., 2022. Millennial and centennial CO2 release from the Southern Ocean during the last deglaciation. Nat. Geosci. 15, 293–299. https://doi.org/10.1038/s41561-022-00910-9.

Zeebe, R.E., Tyrrell, T., 2019. History of carbonate ion concentration over the last 100 million years II: revised calculations and new data. Geochim. Cosmochim. Acta 257, 373–392. https://doi.org/10.1016/j.gca.2019.02.041.

Zhang, Y., de Boer, A.M., Lunt, D.J., Hutchinson, D.K., Ross, P., van de Flierdt, T., Sexton, P., Coxall, H.K., Steinig, S., Ladant, J.B., Zhu, J., Donnadieu, Y., Zhang, Z., Le Chan, W., Abe-Ouchi, A., Niezgodzki, I., Lohmann, G., Knorr, G., Poulsen, C.J., Huber, M., 2022. Early Eocene Ocean Meridional Overturning Circulation: the Roles of Atmospheric Forcing and Strait Geometry. Paleoceanogr. Paleoclimatol. 37. https://doi.org/10.1029/2021PA004329.

Locarnini, R.A., Mishonov, A.V., Antonov, J.I., Boyer, T.P., Garcia, H.E., Baranova, O.K., et al., 2010. World Ocean Atlas 2009, Volume 1: Temperature, 1. World Ocean Atlas 2009, p. 184 (Temperature.).

Garcia, H.E., Locarnini, R.A., Boyer, T.P., Antonov, J.I., Zweng, M.M., Baranova, O.K., Johnson, D.R., 2010. World Ocean Atlas 2009, Volume 4: Nutrients (phosphate, nitrate, silicate), 4. World Ocean Atlas 2009.

Antonov, J.I., D. Seidov, T.P., Boyer, R.A., Locarnini, A.V., Mishonov, H.E., Garcia, O.K., et al., 2010. World Ocean Atlas 2009, Volume 2: Salinity, 2. World Ocean Atlas 2009, p. 184.